

# Evaluating and Mitigating Adversarial Vulnerabilities in Vision Foundation Models

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**Abstract**—Looking at the performance of foundation models, such as Microsoft’s Florence-2, we find that they can achieve state-of-the-art zero-shot results in vision-language tasks. However, the resilience of these models to adversarial attacks has not been thoroughly studied. We present the adversarial evaluation of Florence-2-Base on the COCO object detection benchmark and see a severe drop in mAP, from 0.297 to 0.226 with FGSM (23.9% drop) and 0.166 with PGD (44.1% drop). To mitigate this, we develop a modular, inference-time defense pipeline that fuses lightweight input operations (JPEG compression, prompt/image ensemble) and feature-level functions (adaptive Gaussian noise, spatial smoothing, quantization & channel mixing, pixel dropout). Our approach recovers up to 0.246 mAP under FGSM (recovering 28.2% of the FGSM loss) and 0.199 mAP under PGD (recovering 25.2% of the PGD loss). Our flag-driven framework is transparent, requires no model retraining, and represents an initial, practical step toward improving robustness of prompt-based Vision Foundation Models (VFM) under standard attacks.

**Index Terms**—Vision Language Models, Vision Foundation Models, Fast Gradient Sign Method, Projected Gradient Descent

## I. INTRODUCTION

When Vision Foundation Models (VFM) like Florence-2 [1], Come to mind, they showcase a paradigm shift towards versatile AI systems in the world of AI. Built on the massive FLD-5B dataset [1], these models use a unified sequence-to-sequence architecture that can take on a multitude of tasks such as captioning, object detection and visual grounding, and can cut down on the need for custom components [2]. Florence-2, in particular, is at the forefront of linking spatial hierarchies and semantic precision (Figure 1), and offers impressive results with its zero-shot and fine-tuning capabilities.

Despite VFMs capabilities in various activities, their susceptibility to adversarial attacks remains a critical concern for real world deployment in safety sensitive domains [3], [4]. This is particularly true in robotics, where generalist policies and robot foundation models increasingly rely on Vision-Language Models (VLMs). Their primary implication is to interpret complex scenes and instructions for systematic control, yet their robustness remains a challenge for safe operation [5]. These attacks involve subtle input perturbations designed to cause model failures. Adversarial examples often reveal the properties of the underlying model rather than being mere bugs [6]. The original Florence-2 study [1] did

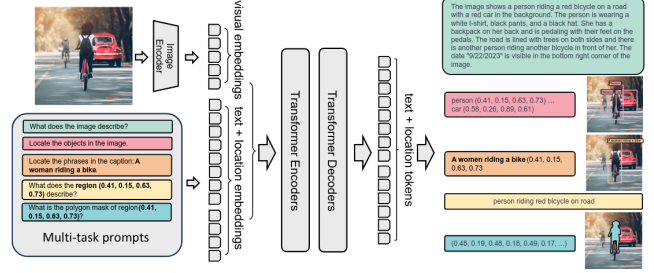


Fig. 1. Florence-2 Model Architecture illustrating the vision encoder (DaViT), multi-modal encoder-decoder (BART-based), visual/text embeddings, and handling of multi-task prompts for text-based output generation. (Image adapted from [1]).

not include robustness evaluations. Potential vulnerabilities identified include a lack of testing against standard adversarial attacks (FGSM, PGD), no proposed defense mechanisms, possible weaknesses of the patch-based tokenization used in its Dual Attention Vision Transformer (DaViT) encoder [7], and uncertain stability under input corruptions or prompt variations in adversarial settings.

## II. RELATED WORK

### A. VFMs

Pretraining large models on extensive datasets is central to modern AI. In vision, models like CLIP [8] and Flamingo [2] leverage vision-language pretraining. Florence-2 [1] employs a unified seq2seq architecture trained on the massive FLD-5B dataset, using components like DaViT [7] for vision encoding.

### B. Adversarial Attacks in Computer Vision

Attacks exploit model vulnerabilities via small input changes [3]. White-box attacks assume full model knowledge. Fast Gradient Sign Method (FGSM) [3] uses a single gradient sign step ( $L_\infty$  norm  $\epsilon$ ); it is fast but often less effective, essentially pushing the input slightly in the direction that maximizes the loss based on the gradient’s sign. Projected Gradient Descent (PGD) [4], uses multiple iterative gradient steps (size  $\alpha$ ) with projection on  $L_\infty$   $\epsilon$ -ball. It refines the adversarial perturbation iteratively to find the potentially stronger adversarial sample within the allowed parameters. We use both to evaluate Florence-2. Adversarial examples can reveal inherent model feature dependencies [6].

### C. Adversarial Defenses

Mitigation strategies include **Adversarial Training**, an often effective but costly method that involves training on attacked images [4], and various other defensive approaches combining methods like innovative preprocessing [9], which include **Input Transformation** methods that preprocess inputs to mitigate perturbations; they are lightweight but potentially bypassable & **Feature-Level Defenses** modify internal activations, introduce randomness, or combine predictions. Our approach is an inference-time pipeline combining input and feature transformations with Prompt Engineering [10] to influence model interaction.

## III. METHODOLOGY

### A. Florence-2 Base Model Overview

We focus on Florence-2-Base (232M parameters) [1]. Its architecture is depicted in Figure 1. It features a **Vision Encoder** (DaViT [7]) that processes an image  $I \in \mathbb{R}^{H \times W \times 3}$  into embeddings  $V$ ; a Bidirectional and Auto-Regressive Transformer (BART)-based **Encoder-Decoder Transformer** that takes projected visual embeddings  $V'$  and prompt embeddings  $T_{prompt}$  to generate an output sequence  $Y$ ; and task specification via text prompts (e.g., ‘OD’ - Object Detection). The model is trained with a cross-entropy loss  $\mathcal{L}$  on the massive **FLD-5B dataset**, which contains 126M images with 5.4B diverse annotations, enabling its strong multitask and generalization capabilities [1].

### B. Adversarial Attack Generation

We implement white-box FGSM and PGD ( $L_\infty$ ) attacks targeting the ‘OD’ task loss  $\mathcal{L}$ . The goal is to find a perturbation  $\delta$  such that  $\|\delta\|_\infty \leq \epsilon$  that maximizes the model’s loss  $\mathcal{L}(\theta, x + \delta, y_{target})$ , effectively degrading detection performance. Here,  $x$  is the normalized input tensor and  $y_{target}$  is the target sequence (e.g., model’s clean prediction).

1) *FGSM Implementation*: FGSM [3] calculates the optimal single-step perturbation under the  $L_\infty$  constraint by moving in the direction of the gradient’s sign:

$$\delta = \epsilon \cdot \text{sign}(\nabla_x \mathcal{L}(\theta, x, y_{target})) \quad (1)$$

This perturbation  $\delta$  is added to the original input  $x$ , and the result is clipped to maintain valid pixel values. It’s a fast method to probe immediate gradient vulnerabilities.

2) *PGD Implementation*: PGD [4] refines the perturbation iteratively over  $N$  steps to find a stronger adversarial example within the  $\epsilon$ -ball:

$$x^{t+1} = \Pi_{\mathcal{B}_\epsilon(x)} (x^t + \alpha \cdot \text{sign}(\nabla_{x^t} \mathcal{L}(\theta, x^t, y_{target}))) \quad (2)$$

At each step  $t$ , it takes a small step  $\alpha$  in the gradient sign direction, then projects (II) the total perturbation ( $x^{t+1} - x$ ) back into the  $L_\infty$  ball of radius  $\epsilon$  centered at the original input  $x$ . The result is also clipped to the valid range.

### C. Proposed Multi-Layered Defense Framework

Our defense combines input and feature-level strategies during inference. The workflow is in Figure 2.

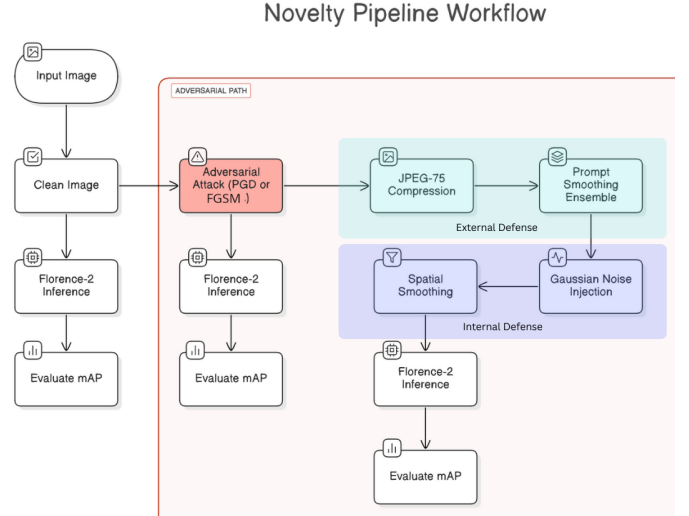


Fig. 2. Workflow diagram of the proposed multi-layered defense pipeline, illustrating the sequence of steps created for adversarial attack and defenses applied during inference. The clean path is shown for comparison.

1) *External (Input-Level) Defenses*: Our external defenses include **JPEG Compression** (quality=75) applied to the input  $I$  to disrupt high-frequency artifacts, and a **Prompt Smoothing Ensemble (PSE)**. The PSE runs multiple inferences with varied prompts ([‘OD’, ‘OD’, ‘Detect...’, ‘Locate...’]) and mild image augmentations (random resize/JPEG), then aggregates weighted detections and applies Non-Maximum Suppression (NMS) (Intersection over Union (IoU)=0.45) for consensus.

2) *Internal (Feature-Level) Defenses*: Applied sequentially to the normalized tensor  $x$ , these defenses start with **Adaptive Gaussian Noise**, adding  $\mathcal{N}(0, \sigma^2)$  noise where  $\sigma \propto \text{mean}(|x|)$  with  $\sigma_{base} \approx 0.06$ . Next, **Adaptive Spatial Smoothing** uses a 5x5 kernel to blend average/median-like filtering based on gradient magnitude, blurring noise while preserving edges. Finally, **Quantization & Channel Mixing** rounds values ( $q_{step} = 0.1$ ), mixes RGB channels, and applies 2% pixel dropout to further discretize the input.

## IV. EXPERIMENTS

### A. Dataset and Metrics

Our primary dataset for evaluating object detection effectiveness is the **COCO 2017 validation set**. It contains 5,000 images with 80 distinct object categories, offering a challenging benchmark for general object detection capabilities.

We report performance using the standard COCO evaluation metrics, calculated using the official ‘pycocotools’: **mAP (mAP@[.5:.95])**: The primary metric, representing the mean Average Precision (AP) computed over IoU thresholds from 0.50 to 0.95 (step 0.05).

### B. Implementation Details

We used the Florence-2-Base model (232M parameters) from Hugging Face Hub (microsoft/Florence-2-base; rev

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**Algorithm 1** Adversarial Detection with Defense Logic (compact)

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**Require:** Image  $I$ , Model  $M$ , Processor  $P$ , flags  $A, D$ , Attack  $Att, \theta_{att}$   
**Ensure:** Final detections  $Det_{final}$

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1:  $I_{in} \leftarrow I$ 
2: if  $A$  then
3:    $I_{preempt} \leftarrow I$ 
4:   if  $D$  then  $I_{preempt} \leftarrow \text{ApplyJPEG}(I, 75)$ 
5:    $I_{in} \leftarrow \text{GenerateAdversarial}(I_{preempt}, M, P, Att, \theta_{att})$ 
6:  $Det_{combined} \leftarrow \emptyset$ 
7: if  $A \wedge D$  then ▷ External defense
8:    $Det_{combined} \leftarrow \text{PromptSmoothingEnsemble}(I_{in}, M, P) \cup Det_{combined}$ 
9:  $x, t_{emb} \leftarrow \text{PreprocessForInference}(I_{in}, P, 'OD')$ 
10: if  $A \wedge D$  then ▷ Internal defenses
11:    $x \leftarrow \text{ApplyInternalNoise}(x, \sigma_{base}=0.06)$ 
12:    $x \leftarrow \text{ApplyInternalSmoothing}(x, \text{kernel}=5)$ 
13:    $x \leftarrow \text{ApplyInternalQuantMixDropout}(x, q_{step}=0.1)$ 
14: with torch.no_grad() :  $gen\_ids \leftarrow M.generate(t_{emb}, x);$   

    $parsed \leftarrow \text{PostProcess}(gen\_ids, P, I_{in}.size)$ 
15:  $Det_{smain} \leftarrow \text{ExtractAndFormatDetections}(parsed, I.size)$ 
16:  $Det_{combined} \leftarrow Det_{combined} \cup Det_{smain}$ 
17:  $Det_{final} \leftarrow \text{ApplyFinalNMS}(Det_{combined}, iou\_thr=0.5)$ 
18: return  $Det_{final}$ 

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TABLE I  
ADVERSARIAL ATTACK ON COCO VAL2017 WITHOUT DEFENSE.

| Condition                         | mAP@[.5:.95] | AP50  | AP75  | mAP Drop vs. Clean |
|-----------------------------------|--------------|-------|-------|--------------------|
| Clean (Reproduced)                | 0.297        | 0.402 | 0.315 | -                  |
| FGSM ( $\epsilon = 0.03$ )        | 0.226        | 0.322 | 0.238 | 0.071 (23.9%)      |
| PGD ( $\epsilon = 0.03, N = 10$ ) | 0.166        | 0.274 | 0.165 | 0.131 (44.1%)      |

refs/pr/26). We consider three different scenarios in our experiments: 1) clean inference, 2) attacked inference without defense and 3) attacked inference with our proposed defense. Our clean inference run yielded an  $\text{mAP}@[.5:.95]=0.297$ , with a total computation time of about 28 minutes with batch size 1 for 5,000 images on RTX 3060.

## V. RESULTS

We present quantitative results on the COCO 2017 validation set. Figures 3 and 4 visualize typical adversarial perturbations generated by our implemented attacks.

### A. Impact of Adversarial Attacks (No Defense)

Table I shows the performance drop under FGSM ( $\epsilon = 0.03$ ) and PGD ( $\epsilon = 0.03$ ) without defenses, using our best clean run ( $\text{mAP}=0.297$ ) as the baseline. The vulnerability is clear, with FGSM causing a  $\approx 23.9\%$  mAP drop and the stronger PGD attack causing a substantial  $\approx 44.1\%$  drop.

### B. Effectiveness of Proposed Defense Framework

Table III presents the results from dedicated defense runs. Our defense strategy yields significant recovery:

- In **FGSM** ( $\epsilon = 0.03$ ), the defense improves mAP from 0.226 to 0.246, recovering 28.2% of performance.
- In **PGD** ( $\epsilon = 0.03$ ), the defense improves mAP from 0.166 to 0.199, recovering 25.2% of performance.



Fig. 3. Example of FGSM attack ( $\epsilon = 0.03$ ). Left: Original. Middle: Adversarial. Right: Magnified Difference (x10) showing high-frequency pattern.



Fig. 4. Example of PGD attack ( $\epsilon = 0.03, N=10$ ). Left: Original. Middle: Adversarial. Right: Magnified Difference (x10) showing distributed pattern.

TABLE II  
ABLATION STUDY OF DEFENSE COMPONENTS UNDER PGD ( $\epsilon = 0.03$ ).

| Defense Configuration | mAP@[.5:.95] | $\Delta\text{mAP}$ | Rel. Recovery (%) |
|-----------------------|--------------|--------------------|-------------------|
| External Only         | 0.178        | +0.012             | 8.8               |
| Internal Only         | 0.187        | +0.021             | 16.4              |
| Full Pipeline         | <b>0.199</b> | <b>+0.033</b>      | <b>25.2</b>       |

These results indicate that the defense improves performance under the evaluated attack settings, though broader verification (stronger/adaptive attacks and cross-model tests) remains future work.

Because the defense includes non-differentiable/stochastic components (JPEG, quantization, dropout), gradient-masking cannot be fully ruled out; BPDA/EOT-based adaptive evaluations remain future work.

### C. Ablation Study

Table II validates the stage-level ablation under PGD ( $\epsilon = 0.03$ ): while External (JPEG, PSE) and Internal (Noise, Smoothing) defenses individually recover +0.012 and +0.021 mAP over the 0.166 baseline, the full framework achieves the peak recovery (+0.033 mAP), confirming that combining input and feature-level strategies yields complementary effectiveness.

## VI. DISCUSSION

In this paper, we present a practical empirical analysis of the robustness of Florence-2-Base against object detection and show its weaknesses, as well as demonstrate the feasibility of a practical defense pipeline without finetuning.

**Adversarial Susceptibility of Florence-2:** The drop in mAP values indicates that, similar to other models trained at a large scale, Florence-2 is susceptible to adversarial attacks, warranting the need to consider robustness prior to deployment. The reasons could be due to model capacity or architecture-related factors, such as patch tokenization [7].

**Defense Mechanism Analysis:** The stable absolute mAP recovery demonstrates the effectiveness of our multilayered

TABLE III  
EFFECTIVENESS OF THE PROPOSED MULTI-LAYERED DEFENSE AGAINST ADVERSARIAL ATTACKS ON COCO val2017 OBJECT DETECTION.

| Attack Type                | Condition (Clean AP baseline: 0.297) | COCO Metrics |              |              | Performance Recovery vs. Clean |               |                   |
|----------------------------|--------------------------------------|--------------|--------------|--------------|--------------------------------|---------------|-------------------|
|                            |                                      | mAP@[.5:.95] | AP50         | AP75         | mAP Drop                       | Abs. Recovery | Rel. Recovery (%) |
| FGSM ( $\epsilon = 0.03$ ) | Clean (Defense Run Baseline)         | 0.297        | 0.402        | 0.315        | -                              | -             | -                 |
|                            | Adversarial (No Defense)             | 0.226        | 0.322        | 0.238        | 0.071 (23.9%)                  | -             | -                 |
|                            | Adversarial (With Defense)           | <b>0.246</b> | <b>0.345</b> | <b>0.258</b> | 0.051 (17.2%)                  | <b>+0.020</b> | <b>28.2%</b>      |
| PGD ( $\epsilon = 0.03$ )  | Clean (Defense Run Baseline)         | 0.297        | 0.402        | 0.315        | -                              | -             | -                 |
|                            | Adversarial (No Defense)             | 0.166        | 0.274        | 0.165        | 0.131 (44.1%)                  | -             | -                 |
|                            | Adversarial (With Defense)           | <b>0.199</b> | <b>0.298</b> | <b>0.202</b> | 0.098 (33.0%)                  | <b>+0.033</b> | <b>25.2%</b>      |

scheme. The defense exhibits good recovery against both FGSM (28.2% relative) and PGD (25.2% relative), indicating its effectiveness against both single-step and stronger iterative attacks. This synergy provides a notable boost in effectiveness without the need for costly model retraining.

#### A. Broader Impacts and Ethical Considerations

When powerful vision and foundation models like Florence-2 are found to be vulnerable to adversarial attacks, it poses significant risks in the real world, and has serious implications in safety-critical applications such as autonomous driving, medical image analysis and the control of dexterous robots [5]. If a well-crafted attack causes misdetection or misclassifications in these areas, the results could be devastating. Visual recognition systems can be bypassed, as dangerous content could be constructed by exploiting areas which use visual recognitions. Also, it is worrisome to be able to manipulate the outputs of generative models, and even though this isn't always done through the model itself, it can be done through feeding it false data, going through the detectors that feed into it. Because the failure of these models has been shown to lead to a lot of harm, it's critical to thoroughly test the robustness of vision and foundation models and build effective defenses, and our work here contributes by showing risks for Florence-2 in object recognition and shows a workable plan to defend against these risks, yet further study of more resilient models, more forceful attacks, guaranteed defenses and the social implications of potential failures still has to be done.

#### B. Limitations

Our evaluation focuses on standard white-box FGSM and PGD attacks on Florence-2-Base using COCO val2017; stronger adaptive attacks (e.g., AutoAttack, BPDA/EOT), black-box transfer, and physical-world perturbations are outside the current scope. Since our defense employs non-differentiable and stochastic operations (JPEG, quantization), we acknowledge that gradient masking cannot be fully ruled out against adaptive attackers. Furthermore, our ablation is conducted at the stage level (external vs. internal), and detailed per-class failure analysis is deferred to future work.

### VII. CONCLUSION

This paper evaluated the adversarial robustness of Florence-2-Base in object detection and showed that it is vulnerable to both FGSM and PGD. To mitigate this, we developed

and tested a multilayered inference-time defense framework through input and feature-level manipulation. Our experiments on the COCO val2017 dataset demonstrated substantial mAP recovery, reclaiming over 28% of the performance loss from standard attacks without model retraining. We presented our methodology, findings, and general implications of our work. This project highlights the importance of examining robustness in contemporary VFM models and provides a practical defense strategy to effectively improve models such as Florence-2 for more reliable use.

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